

Scope Over Monotonicity: NPI Licensing with Czech majority

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Introduction

- traditional generalization: **NPIs** are licensed under **downward-entailing operators** (Ladusaw (1979))

1. Some linguistic data escape this rule

(AND the weakened version by von Stechow (1999)) :

- (1)
- a. **Most** men with *any* revolutionary commitments were executed.
 - b. **Half** of those who've *ever* voted are no longer living.
 - c. **Exactly three** people did *any* work *at all*.
 - d. **Usually**, women who've *ever* seen Paris want to go back sometime.

(from Rothschild (2006a), Rothschild (2006b))

2. Experimental evidence remains mixed:

- Szabolcsi et al. (2008): NPIs do not facilitate downward inferences (DI)
- Chemla et al. (2011); Denić et al. (2021) : subjective judgments → DI → licensing

3. An alternative approach:

- Barker (2018):
 - NPIs signal **narrow scope** wrt. licensor
 - licensing = non-entailment between two scope readings
- in linguistic data, both conditions are usually met

- Dočekal and Chumchalová (2025) : double licensing structures
X doesn't doubt that [... NPI ...]:
 - NPI licensing is **domain-sensitive** (Homer (2021))
 - Yet inference patterns remained **uniformly upward-entailing**
- **Problem:** DL environments still involve *monotonic* operators (negation, *doubt*) with complex scope interactions
- **Question:** Can NPIs appear in *empirically non-monotonic* environments?

Solution:

- *Proportional* quantifier (Barwise and Cooper (1981))
- Neither downward- nor upward-entailing
- *většina* = “majority”, “most”
- $\llbracket \textit{většina} \rrbracket(A)(B) = 1$ iff $|A \cap B| > |A \setminus B|$

- “the majority” is not DE:

(2) Context: room with 10 cats, 8 black, 2 white.

a. The majority of cats in this room are black. True

b. The majority of white cats in this room are black.

Contradiction

- but it does license NPIs (variation of Rothschild’s example):

(3) **The majority** of men with *any* revolutionary commitments were executed.

Hypothesis: NPIs are licensed under *většina* while participants simultaneously **reject both DE and UE** inferences - If true → stronger evidence against the DI–NPI link

Experiment

- **Platform:** PCIBex Farm (Zehr and Schwarz (2018))
- Two sections: acceptability task + inference judgment task
- **Hypothesis:** PI *sebemenší* in restrictor of *most*-QP would be judged **acceptable** despite **non-monotonic** environment
- **Task:** Two sections = **acceptability** + **inference judgment**
- **Participants:** two versions based on task ordering
 - Acceptability-first: N=24
 - Inference-first: N=29
- No overlap in samples, no demographic info, option of course credits for Masaryk U. students, anonymization of datasets

Acceptability Task

- 9 items in 1×3 factorial design
- Conditions varied by quantifier in QP above PI:
 - DE: *all*
 - NM: *most*
 - UE: *some*
- 1–5 Likert scale
- Expectation:

all > most > some

Example item:

- (4) *Všichni/Většina/Někteří občané/-ů se sebemenší vírou v Boha chtěli/-a nalezenému dítěti vystrojit křest.*

‘All/Most/Some of the citizens with even the least faith in God wanted to arrange a christening for the found child.’

- Fillers = various sentences of a similar length to the items

Inference Task

- 18 items in 2×3 factorial design
- **Factor I:** we asked about upward/downward entailment
 - UE: *circle* $\rightarrow$ *shape*
 - DE: *circle* $\rightarrow$ *green.circle*
- **Factor II:** the quantifiers in our stimuli (*all/most/some*)
- **Visuals** – three layers of subset relations
 - **shapes:** ○ ○ ○ ○ ○ ○ ♥ ♥ ♥ ♥ ♥ ♥
↑
 - **circles:** ○ ○ ○ ○ ○ ○
↑
 - **green circles:** ○ ○ ○

- BEFORE = Top of the image.
- CLEANUP = Klára places the 12 shapes into two boxes.
- AFTER = Bottom of the image.
- window in one box reveals contour but not color
- Boris knows BEFORE, did not witness CLEANUP, now sees AFTER
- AFTER = Sentence A
Boris infers Sentence B

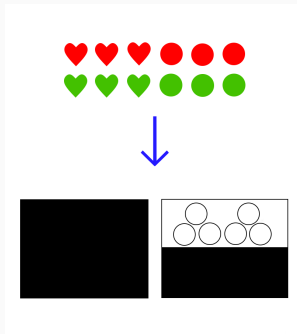


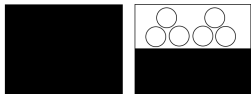
Figure 1: Item 14,
conditions UE-ALL & DE-ALL

Sentence A: All the *circles* are in the box on the right.

Sentence B, UE: All the *shapes* are in the box on the right.

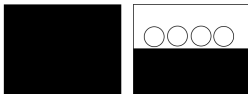
Sentence B, DE: All the *green circles* are in the box on the right.

Did Boris infer B correctly? Click YES or NO!



UE-ALL:
All of the shapes?

DE-ALL:
All of the green circles?



UE-MOST:
Most of the shapes?

DE-MOST:
Most of the green circles?



UE-SOME:
Some of the shapes?

DE-SOME:
Some of the green circles?

(Logically) correct answers:

no

no

yes

yes

no

no

Expectation: The participants will mostly answer correctly.
(We did not tell them there was any “correct” or “incorrect”.)

Results

- Acceptability ratings and inference responses shown in Figure~2
- **Analyses:** Bayesian ordinal regression (cumulative logit link) hierarchical models
- **Tool:** *R* with *brms* (Bürkner (2017))
- Version order showed no credible effect ($|\hat{\beta}| < 0.3$, $BF_{01} > 4$)

Acceptability Ratings & Inference Responses

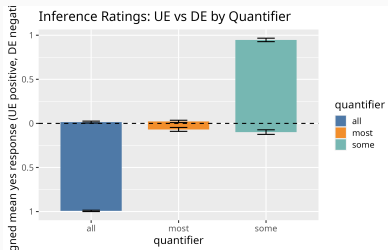


Figure 2: Acceptability ratings with error bars (left) and inference responses (right)

- Bayesian ordinal regression (cumulative logit link) in *R* with *brms* (Bürkner (2017))
- Weakly informative priors
- Random intercepts: by-participant and by-item (maximal models failed to converge, so simplified random effects structure used)
- **Reference level:** *most*

- **Pattern:** Decisive evidence for expected ordering
 - *all* rated credibly **higher** than *most* ($\hat{\beta} = 0.96$, 95% CrI [0.31, 1.60], $BF_{10} = 16.31$: strong evidence)
 - *some* rated credibly **lower** than *most* ($\hat{\beta} = -1.18$, 95% CrI [-1.80, -0.55], $BF_{10} = 104.60$: extreme evidence)
 - *Note: coefficients on cumulative log-odds scale*
- Predicted differences on the Likert scale:
 - *all*: +0.48 points
 - *some*: -0.73 points

Posterior Densities

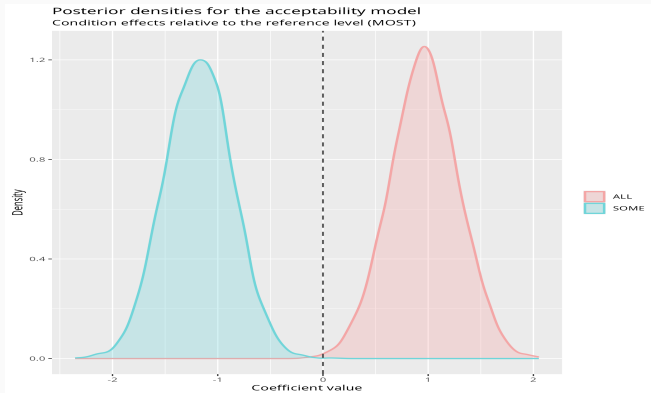


Figure 3: Posterior densities for the acceptability model: Condition effects relative to the reference level (MOST). ALL (red) is positively shifted, SOME (teal) is negatively shifted, both with minimal overlap at zero.

Inference Findings

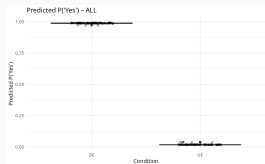
- Compared inference patterns across quantifiers (*all*, *most*, *some*)
- Bayesian logistic regression (hierarchical) in *R* with *brms*
- Weakly informative priors
- Random intercepts: by-participant and by-item (maximal models failed to converge, so simplified random effects structure used)

- **Reference:** *most* as reference level; DE as reference direction
- **Results:**
 - Extreme evidence for DE under *all* ($\hat{\beta}_{\text{all}} = 5.71$, 95% CrI [4.78, 6.76], $\text{BF}_{10} = 3.41 \times 10^{10}$)
 - plus interaction: $\hat{\beta}_{\text{UE:all}} = -5.55$, 95% CrI [-7.02, -4.09], $\text{BF}_{10} = 1.04 \times 10^7$
 - Extreme evidence for UE under *some* ($\hat{\beta}_{\text{UE:some}} = 5.99$, 95% CrI [4.77, 7.31], $\text{BF}_{10} = 4.08 \times 10^{11}$)
 - **Critically:** *most* showed **no evidence** for either direction
- again coefficients on the log-odds (logit) scale, not on the probability scale

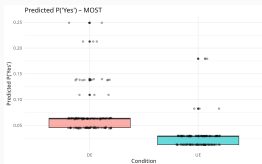
- posterior summaries on the probability scale:

condition	median	lo95	hi95
DE-most	0.080	0.039	0.144
DE-all	0.963	0.919	0.985
DE-some	0.114	0.061	0.191
UE-most	0.025	0.009	0.062
UE-all	0.031	0.010	0.072
UE-some	0.938	0.882	0.973

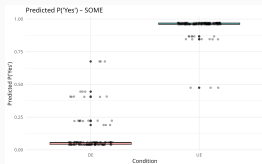
1. *all*: DE is much higher than UE (0.963 vs 0.031), so very strong DE pattern.
2. *some*: UE is much higher than DE (0.938 vs 0.114), so very strong UE pattern.
3. *most*: both are low (0.080 and 0.025), so mostly “no” in both directions; only a weak DE tilt, not a clean DE/UE classification.



(a) ALL



(b) MOST



(c) SOME

Figure 4: Predicted probability distributions by quantifier.

- **Dissociation confirmed:** *most* is neither downward nor upward entailing
- Despite this, *most* licenses NPIs at **intermediate levels**
- **Acceptability gradient:** *all* (+0.48 points), *most* (baseline), *some* (-0.73 points)

Discussion

- **Critical dissociation:** Non-monotonic *most* licenses NPIs at intermediate levels ($\mu = 3.95$) despite showing **no monotonic inference pattern** (DE: 6.8%; UE: 2.3%)
- Directly challenges Ladusaw's (1979) hypothesis: **downward entailment is NOT a synchronic licensing condition** for NPIs

Why This Matters

- Unlike DL environments (which involve monotonic operators with complex scope interactions)
- *most* is **genuinely non-monotonic** in its restrictor
- Participants robustly rejected **both** DE and UE inferences at near-floor levels (the probability of inferring DE under *most* = 0.08; UE = 0.025)
- Confirms *most* as a non-DE/non-UE environment

- Decisive monotonicity effects:
 - **Extreme** DE evidence under *all* (the probability of inferring DE under *all* = 0.963; $BF_{10} = 3.41 \times 10^{10}$)
 - **Extreme** UE evidence under *some* (the probability of inferring UE under *some* = 0.938; $BF_{10} = 4.08 \times 10^{11}$)
 - *most* yielded **no credible preference**
- Acceptability gradient mirrors monotonicity landscape:
 - Maximal DE licensing → highest acceptability
 - Genuine non-monotonicity → intermediate acceptability
 - Maximal UE → lowest acceptability

- Pattern **incompatible** with Ladusaw's synchronic DE requirement
- **Aligns** with Barker (2018)'s scope-marking view:
 - NPIs signal **narrow scope** with respect to their licenser
 - Licensed when wide scope does **not** entail narrow scope

- Formalization of the example condition:

- (5) [_F The majority of citizens with even the least faith in God]
 [_P wanted to arrange a christening for the found child.]
- $\exists d [\text{většina}_x(F(x, d) \rightarrow P(x))] > \text{(wide scope)}$
 - $\text{většina}_x((\exists d F(x, d)) \rightarrow P(x)) > \text{(narrow scope)}$

From $\exists d [\text{většina}_x(F(x, d) \rightarrow P(x))]$ (there is some degree of faith such that most people satisfy P) one cannot infer $\text{většina}_x((\exists d F(x, d)) \rightarrow P(x))$ (most people with some degree of faith satisfy P) since there may be a threshold of d that guarantees P (first formula true, second false)

Scope Entailments & NPI Licensing

Two scope readings for *sebemenší* (“even the least”):

- **Wide d:** $\exists d [Q_x (F(x, d) \dots P(x))]$
“There is threshold *d* such that *Q* people with $\geq d$ faith...”
- **Narrow $\exists d$:** $Q_x ((\exists d F(x, d)) \dots P(x))$
“*Q* people with **any** faith...”

Does Wide (a) entail Narrow (b)?

Quantifier	Entailment	Result	Acceptability
ALL (DE)	NO ($\not\models$)	✓ NPI Licensed	Highest
MOST (NM)	NO ($\not\models$)	✓ NPI Licensed	Intermediate
SOME (UE)	YES ($\models$)	× NOT Licensed	Lowest

*Licensing requires **non-entailment** between scope readings*

Conclusion

- Under *most*, the NPI takes narrow scope within the restrictor (*sebemenší vírou*, “least faith”)
- Yields an interpretable configuration despite non-monotonicity
- Results replicate Szabolcsi et al. (2008)’s null finding
- Extends this to a theoretically critical case: an **empirically non-monotonic** quantifier
- Diachronic note: NPIs may have grammaticalized in DE environments (Herburger (2023)), but DE is **not synchronically necessary** for licensing

Thank you for your attention!



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